1. Introduction
Retrieval-Augmented Generation (RAG) is an AI framework for improving the quality of LLM responses by groun

2. Methodology
We propose a lightweight architecture using local 7B quantized models and CPU-based embedding systems. Th

3. Results
Evaluation on RTX 4050 shows low thermal footprint and 45 tokens per second inference speed with zero halluc

4. Conclusion
Local RAG enables private, highly specialized domain expertise on consumer hardware.